

Multi-Agent Reinforcement Learning & Agentic AI

Deep Reinforcement Learning

This lecture is based on

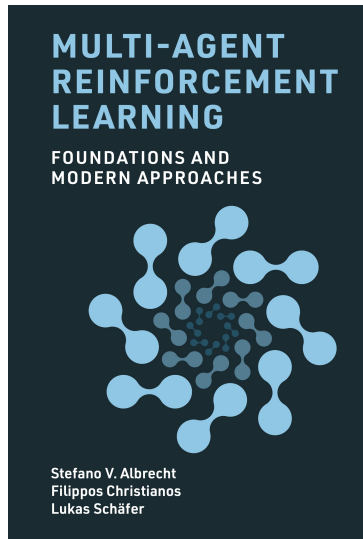
**Multi-Agent Reinforcement Learning:
Foundations and Modern Approaches**

by Stefano V. Albrecht, Filippos Christianos and
Lukas Schäfer

MIT Press, 2024

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www.marl-book.com



Lecture Outline

- Deep-Q learning
- Moving target problem
- Addressing correlations in consecutive experiences
- Policy gradient algorithms

Deep Q-Learning

Recap

Yesterday, we looked at 3 RL algorithms for solving an MDP to maximise the sum of discounted rewards:

- Policy iteration: model-based, policy-based.
- Value iteration: model-based, value-based.
- Q-learning: model free, value-based.

Recap: Q-Learning

Reminder

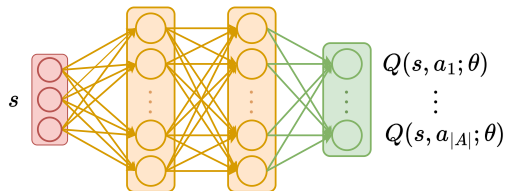
The Bellman update rule in tabular Q-learning is:

$$\underbrace{Q(s^t, a^t)}_{\text{New Q-value}} \leftarrow \underbrace{Q(s^t, a^t)}_{\text{Old Q-value}} + \underbrace{\alpha}_{\text{Learning rate}} \left(\underbrace{r^t}_{\text{Immediate reward}} + \underbrace{\gamma \max_{a'} Q(s^{t+1}, a') - Q(s^t, a^t)}_{\text{TD target}} \right)$$

TD error

Deep Q-Learning

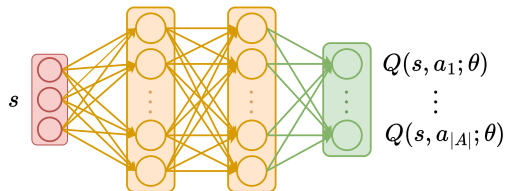
For **deep** Q-learning, we use a neural network to approximate the Q function.



- To train this we could define a loss function
$$\mathcal{L}(\theta) = (y^t - Q(s^t, a^t; \theta))^2$$
- But unlike supervised learning, we are not given y^t beforehand

Deep Q-Learning

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- To train this we could define a loss function $\mathcal{L}(\theta) = (y^t - Q(s^t, a^t; \theta))^2$
- But unlike supervised learning, we are not given y^t beforehand
- We can use the Q-learning update rule to define our y^t

$$y^t = \begin{cases} r^t & \text{if } s^{t+1} \text{ is terminal} \\ r^t + \gamma \max_{a'} Q(s^{t+1}, a'; \theta) & \text{otherwise} \end{cases}$$

Naive Deep Q-Learning Pseudo-Code

Algorithm Deep Q-learning

```
1: Initialize value network  $Q$  with random parameters  $\theta$ 
2: for every episode do
3:   for time step  $t = 0, 1, 2, \dots$  do
4:     Observe current state  $s^t$ 
5:     With probability  $\epsilon$ : choose random action  $a^t \in A$ 
6:     Otherwise: choose  $a^t \in \arg \max_a Q(s^t, a; \theta)$ 
7:     Apply action  $a^t$ ; observe reward  $r^t$  and next state  $s^{t+1}$ 
8:     if  $s^{t+1}$  is terminal then
9:       Target  $y^t \leftarrow r^t$ 
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This naive application of neural networks to RL algorithms has some problems.

The Moving Target Problem

Problem

Moving target problem arises from the bootstrapped targets:

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- As targets y^t depend on θ , any update to θ changes the target
⇒ This non-stationarity of the targets makes it difficult to learn optimal θ

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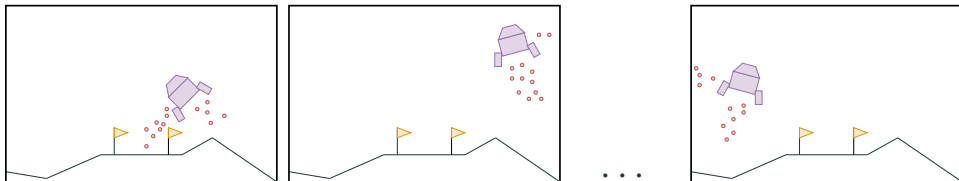
- Value function with NNs generalize value estimates across inputs
- As targets y^t depend on θ , any update to θ changes the target
 $\Rightarrow$ This non-stationarity of the targets makes it difficult to learn optimal θ

Solution

One solution is to use a **target network** with parameters $\bar{\theta}$ that are updated less often than our Q network's parameters θ

Correlation of Consecutive Experiences

- Most ML algorithms using NNs assume i.i.d. data
- In RL, we collect data by interacting with an MDP with $s^{t+1} \sim \mathcal{T}(s^t, a^t) \rightarrow$ **not** i.i.d



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This correlated data can lead to **overfitting** of the value function to recent experiences, and result in **catastrophic forgetting** of previously learned estimates.

Solution

We train on samples of previous experiences stored in a **replay buffer**.

Deep Q-Networks

Deep Q-networks (DQN) is a foundational deep RL algorithm based on Q-learning.

- Tabular value function $\longrightarrow$ neural network
- Moving target problem $\longrightarrow$ target networks
- Correlated experiences $\longrightarrow$ replay buffer

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Target networks:

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- Update the "main" value network parameters θ by minimizing the DQL loss $\mathcal{L}(\theta)$
- Update the **target network** parameters in regular intervals $\bar{\theta} \leftarrow \theta$

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Replay buffers:

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Note

Replay buffers can only be used for **off-policy** algorithms. We use experiences collected from previous (different) policies, which change as we update θ .

DQN Pseudo Code

Algorithm Deep Q-networks (DQN)

```
1: Initialize value network  $Q$  with random parameters  $\theta$ 
2: Initialize target network with parameters  $\bar{\theta} = \theta$ 
3: Initialize an empty replay buffer  $\mathcal{D} = \{\}$ 
4: for every episode do
5:   for time step  $t = 0, 1, 2, \dots$  do
6:     Observe current state  $s^t$ 
7:     With probability  $\epsilon$ : choose random action  $a^t \in A$ 
8:     Otherwise: choose  $a^t \in \arg \max_a Q(s^t, a; \theta)$ 
9:     Apply action  $a^t$ ; observe reward  $r^t$  and next state  $s^{t+1}$ 
10:    Store transition  $(s^t, a^t, r^t, s^{t+1})$  in replay buffer  $\mathcal{D}$ 
11:    Sample random mini-batch of  $B$  transitions  $(s^k, a^k, r^k, s^{k+1})$  from  $\mathcal{D}$ 
12:    if  $s^{k+1}$  is terminal then
13:      Targets  $y^k \leftarrow r^k$ 
14:    else
15:      Targets  $y^k \leftarrow r^k + \gamma \max_{a'} Q(s^{k+1}, a'; \bar{\theta})$ 
16:    Loss  $\mathcal{L}(\theta) \leftarrow \frac{1}{B} \sum_{k=1}^B (y^k - Q(s^k, a^k; \theta))^2$ 
17:    Update parameters  $\theta$  by minimising the loss  $\mathcal{L}(\theta)$ 
18:    In a set interval, update target network parameters  $\bar{\theta}$ 
```

Problem

DQN tends to **overestimate** values with 1-step targets

$$y^k \leftarrow r^k + \gamma \max_{a'} Q(s^{k+1}, a'; \bar{\theta})$$

- Using the max operator, select the maximum value estimate for the target
- Since our value estimates do not necessarily reflect the true value function, the max operation will likely select an overestimated action-value estimate
- This can slow down the convergence of the algorithm as the agent spends too much time exploring states with overestimated values

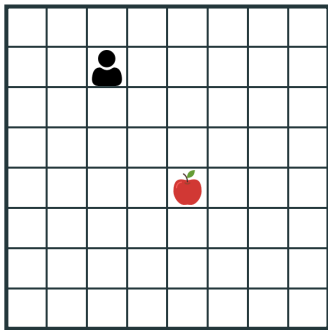
Solution

Double Q-learning reduces this overestimation bias by decoupling the action selection from value estimation using separate function approximations.

- This can be achieved with minimal changes in DQN
- DDQN uses the primary Q-network (with parameters θ) to select actions while using the target network (with parameters $\bar{\theta}$) to estimate action values
- The target thus becomes:

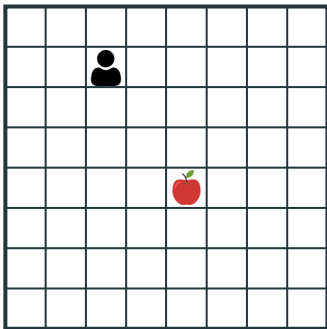
$$y^t = \begin{cases} r^t & \text{if } s^{t+1} \text{ is terminal} \\ r^t + \gamma Q(s^{t+1}, \arg \max_{a'} Q(s^{t+1}, a'; \theta); \bar{\theta}) & \text{otherwise} \end{cases}$$

Deep Q-learning in Practice

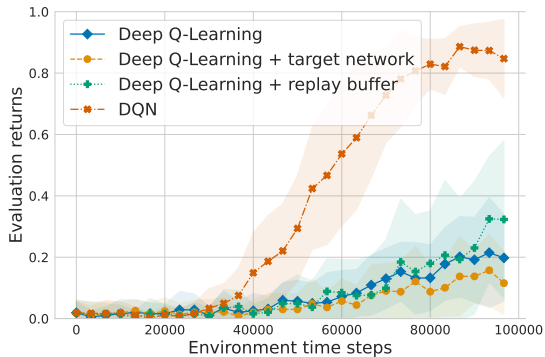


(a) Single-agent level-based foraging environment

Deep Q-learning in Practice

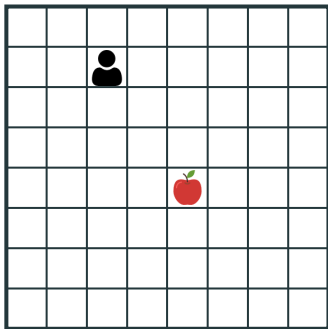


(a) Single-agent level-based foraging environment

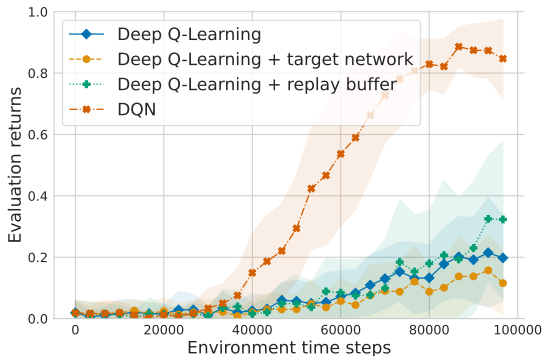


(b) Learning curves

Deep Q-learning in Practice



(a) Single-agent level-based foraging environment



(b) Learning curves

Note that in isolation, neither the addition of **target networks** nor of a **replay buffer** are sufficient to stably train the agent with deep Q-learning in this environment.

Policy Gradient Algorithms

Policy Gradients

We considered a parameterized value function, but we can also directly parameterize the policy π .

- Use a NN for policy π with parameters ϕ
- Policy network receives state s as input and outputs a scalar value for each action
- Scalars $l(s, a)$ represent the preference of the policy to select action a in state s

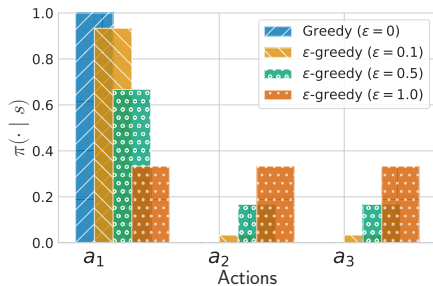
Policy Gradients

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- Policy network receives state s as input and outputs a scalar value for each action
- Scalars $I(s, a)$ represent the preference of the policy to select action a in state s
- Preferences are then transformed into a probability distribution across the action space using a **softmax** function:

$$\pi(a \mid s; \phi) = \frac{e^{I(s,a;\phi)}}{\sum_{a' \in A} e^{I(s,a';\phi)}}$$

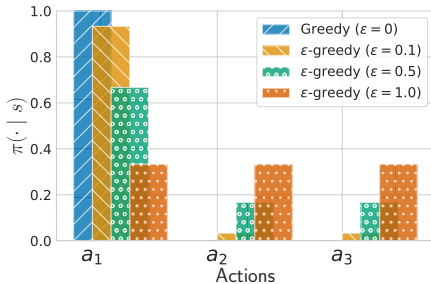
Advantages of Learning a Policy



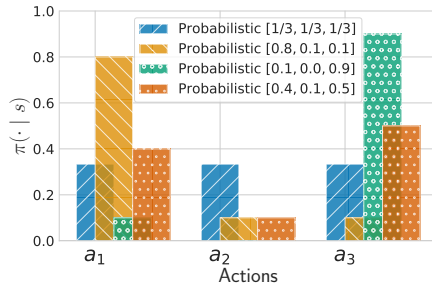
ϵ -greedy policies

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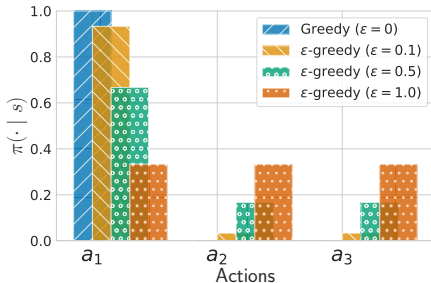
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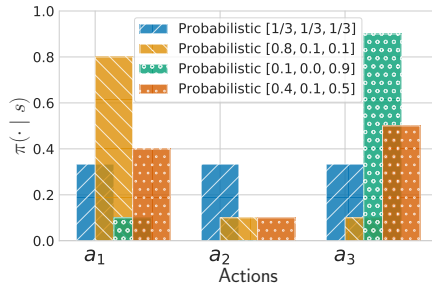
Probabilistic policies

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- Policy gradient algorithms allow us to represent **any** probabilistic policy

Advantages of Learning a Policy



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Probabilistic policies

- ϵ -greedy policies struggle to represent diverse probabilistic policies
- Policy gradient algorithms allow us to represent **any** probabilistic policy
- Policy gradients are also effective for representing **continuous** action spaces

Policy Gradient Theorem

How to update parameters ϕ of the policy? Using the **policy gradient theorem**, we can express the gradient of the performance of a policy with respect to the parameter ϕ of the policy.

$$\nabla_{\phi} J(\phi) \propto \sum_{s \in S} \Pr(s \mid \pi) \sum_{a \in A} Q^{\pi}(s, a) \nabla_{\phi} \pi(a \mid s; \phi)$$

- J is a function measuring the quality of policy π
- $\Pr(s \mid \pi)$ is the state-visitation distribution for policy π
- $Q^{\pi}(s, a)$ is the value for a given action and state under π
- The J function is similar to a loss function, with the difference that we aim to **maximize** rather than minimize it

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Note

The policy gradient theorem assumes that $\Pr(s \mid \pi)$ are given under the currently optimized policy $\pi \rightarrow$ this needs **on-policy** data

Policy Gradient Theorem – Continued

The assumption that $Pr(s \mid \pi)$ is **on-policy** becomes apparent through a simple derivation.

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Policy Gradient Theorem – Continued

Intuition

$$\nabla_{\phi} J(\phi) \propto \mathbb{E}_{a \sim \pi(\cdot | s; \phi)} \left[Q^{\pi}(s, a) \frac{\nabla_{\phi} \pi(a | s; \phi)}{\pi(a | s; \phi)} \right]$$

We can interpret the components of the policy gradient theorem as follows:

- $\nabla_{\phi} \pi(a | s; \phi)$: gradient of ϕ pointing in the direction that most increase the probability of taking action a when in state s

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- $Q^{\pi}(s, a)$: expected returns as a quality measure of taking action a in state s
- $\frac{1}{\pi(a | s; \phi)}$: normalization coefficient to account for varying probabilities of different actions under the policy

REINFORCE: Monte Carlo

To apply the policy gradient theorem, we must find a way to derive expected returns.

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- The algorithm minimizes the following loss for an episodic history $h = \{s^0, a^0, r^0, \dots, s^{T-1}, a^{T-1}, r^{T-1}, s^T\}$:

$$\mathcal{L}(\phi) = -\frac{1}{T} \sum_{t=0}^{T-1} \left(\sum_{\tau=t}^{T-1} \gamma^{\tau-t} r^{\tau} \right) \log \pi(a^t | s^t; \phi)$$

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- Note this loss has a negative sign, as we want to **maximize** expected returns

REINFORCE Pseudocode

Algorithm REINFORCE

- 1: Initialize policy network π with random parameters ϕ
 - 2: **for** every episode **do**
 - 3: **for** time step $t = 0, 1, 2, \dots, T - 1$ **do**
 - 4: Observe current state s^t
 - 5: Sample action $a^t \sim \pi(\cdot \mid s^t; \phi)$
 - 6: Apply action a^t ; observe reward r^t and next state s^{t+1}
 - 7: Loss $\mathcal{L}(\phi) \leftarrow -\frac{1}{T} \sum_{t=0}^{T-1} \left(\sum_{\tau=t}^{T-1} \gamma^{\tau-t} r^{\tau} \right) \log \pi(a^t \mid s^t; \phi)$
 - 8: Update parameters ϕ by minimizing the loss $\mathcal{L}(\phi)$
-

Baseline to Reduce Variance

Problem

High variance of MC estimates causes unstable gradients and training.

Baseline to Reduce Variance

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High variance of MC estimates causes unstable gradients and training.

Solution

- Reduce variance by subtracting a **baseline** from the return estimates
- Common choice of baseline is state-value function $V(s)$ which can be trained to minimize the loss $\mathcal{L}(\theta) = \frac{1}{T} \sum_{t=1}^{T-1} (u(h^t) - V(s^t; \theta))^2$
- The REINFORCE policy loss then becomes

$$\mathcal{L}(\phi) = -\frac{1}{T} \sum_{t=0}^{T-1} (u(h^t) - V(s^t; \theta)) \log \pi(a^t | s^t; \phi)$$

Actor-Critic Algorithms

The policy gradient theorem allows us to optimize our parameterized policy; we still need to approximate expected returns.

- MC methods have high variance and require an entire episode to compute estimates
- **Actor-critic** algorithms aim to reduce variance and update more often by using **bootstrapped return estimates**

$$\begin{aligned}\mathbb{E}_{\pi} [u(h^t) \mid s^t] &= \mathbb{E}_{\pi} [R(s^t, a^t, s^{t+1}) + \gamma u(h^{t+1}) \mid s^t, a^t \sim \pi(\cdot \mid s^t)] \\ &= \mathbb{E}_{\pi} [R(s^t, a^t, s^{t+1}) + \gamma V(s^{t+1}) \mid s^t, a^t \sim \pi(\cdot \mid s^t)]\end{aligned}$$

- We now train a **critic** (value function approximator) alongside the **actor** (parameterized policy) which acts as a baseline

Balancing Bias and Variance

Problem

MC estimates have high **variance** and bootstrapping introduces **bias** since the value function might not yet approximate the true expected returns.

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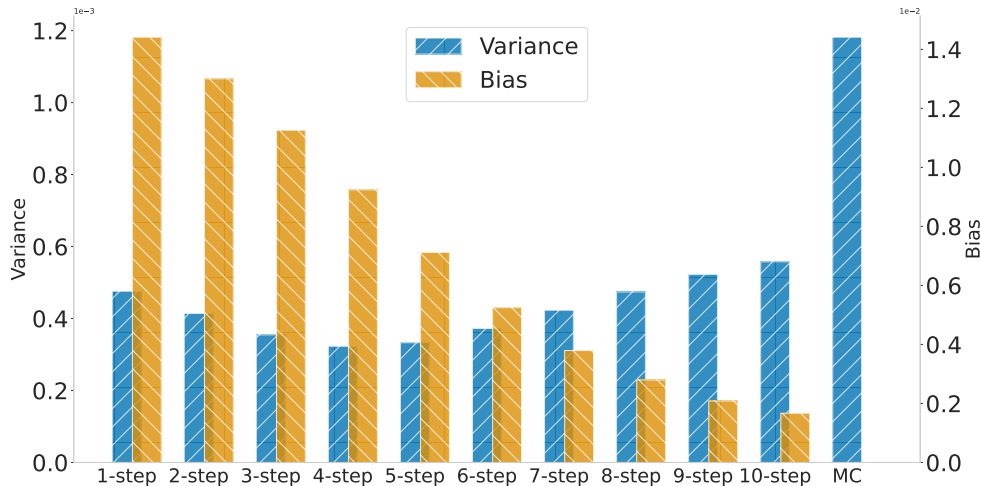
Solution

N-step returns allow us to balance between bias and variance:

$$\mathbb{E}_{\pi}[u(h^t) \mid s^t] = \mathbb{E}_{\pi} \left[\left(\sum_{\tau=0}^{N-1} \gamma^{\tau} \mathcal{R}(s^{t+\tau}, a^{t+\tau}, s^{t+\tau+1}) \right) + \gamma^N V(s^{t+N}) \mid s^t, a^{\tau} \sim \pi(\cdot \mid s^{\tau}) \right]$$

$N = 1 \rightarrow$ one-step bootstrapped returns ... $N = T \rightarrow$ MC returns

Balancing Bias and Variance – Continued



Advantage Actor Critic (A2C)

Advantage actor-critic is a foundation actor-critic algorithm which uses the **advantage** of a policy to guide the policy gradients:

$$Adv^{\pi}(s, a) = Q^{\pi}(s, a) - V^{\pi}(s)$$

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We can compute the advantage using only a state-value function:

$$Adv(s^t, a^t) = Q(s^t, a^t) - V(s^t) = \begin{cases} r^t - V(s^t) & \text{if } s^{t+1} \text{ is terminal} \\ r^t + \gamma V(s^{t+1}) - V(s^t) & \text{otherwise} \end{cases}$$

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- We can also still use N-step returns to reduce bias during the advantage computation

Algorithm A2C

```
1: Initialize actor network  $\pi$  with random parameters  $\phi$ 
2: Initialize critic network  $V$  with random parameters  $\theta$ 
3: for every episode do
4:   for time step  $t = 0, 1, 2, \dots$  do
5:     Observe current state  $s^t$ 
6:     Sample action  $a^t \sim \pi(\cdot \mid s^t; \phi)$ 
7:     Apply action  $a^t$ ; observe reward  $r^t$  and next state  $s^{t+1}$ 
8:     if  $s^{t+1}$  is terminal then
9:       Advantage  $Adv(s^t, a^t) \leftarrow r^t - V(s^t; \theta)$ 
10:      Critic target  $y^t \leftarrow r^t$ 
11:    else
12:      Advantage  $Adv(s^t, a^t) \leftarrow r^t + \gamma V(s^{t+1}; \theta) - V(s^t; \theta)$ 
13:      Critic target  $y^t \leftarrow r^t + \gamma V(s^{t+1}; \theta)$ 
14:      Actor loss  $\mathcal{L}(\phi) \leftarrow -Adv(s^t, a^t) \log \pi(a^t \mid s^t; \phi)$ 
15:      Critic loss  $\mathcal{L}(\theta) \leftarrow (y^t - V(s^t; \theta))^2$ 
16:      Update parameters  $\phi$  by minimizing the actor loss  $\mathcal{L}(\phi)$ 
17:      Update parameters  $\theta$  by minimizing the critic loss  $\mathcal{L}(\theta)$ 
```

Proximal Policy Optimization (PPO)

Problem

Policy gradient methods can cause significant shifts in the policy with a single update, which can worsen the policy!

Solution

Limit the change of the policy in a single update → **trust region** of a policy

Proximal policy optimization (PPO) computes an efficient surrogate objective to limit the change in the policy when executing multiple updates:

$$\mathcal{L}(\phi) = -\min \left(\begin{array}{l} \rho(s^t, a^t) \text{Adv}(s^t, a^t), \\ \text{clip}(\rho(s^t, a^t), 1 - \epsilon, 1 + \epsilon) \text{Adv}(s^t, a^t) \end{array} \right)$$

PPO Surrogate Objective

$$\mathcal{L}(\phi) = -\min \left(\begin{array}{l} \rho(s^t, a^t) \text{Adv}(s^t, a^t), \\ \text{clip}(\rho(s^t, a^t), 1 - \epsilon, 1 + \epsilon) \text{Adv}(s^t, a^t) \end{array} \right)$$

- ρ represents the importance sampling ratio $\rho(s, a) = \frac{\pi(a|s; \phi)}{\pi_\beta(a|s)}$
- π_β represents the behavior policy followed to select action a^t in state s^t
- ϵ is a hyperparameter that determines the allowed change of the policy

PPO Surrogate Objective

$$\mathcal{L}(\phi) = -\min \left(\begin{array}{l} \rho(s^t, a^t) Adv(s^t, a^t), \\ \text{clip}(\rho(s^t, a^t), 1 - \epsilon, 1 + \epsilon) Adv(s^t, a^t) \end{array} \right)$$

- ρ represents the importance sampling ratio $\rho(s, a) = \frac{\pi(a|s; \phi)}{\pi_\beta(a|s)}$
- π_β represents the behavior policy followed to select action a^t in state s^t
- ϵ is a hyperparameter that determines the allowed change of the policy

Importance sampling ratios serve multiple purposes:

- Correct for differences in data distributions of π_β and π
- Measure of divergence between π_β and $\pi \rightarrow$ can be clipped to limit divergence

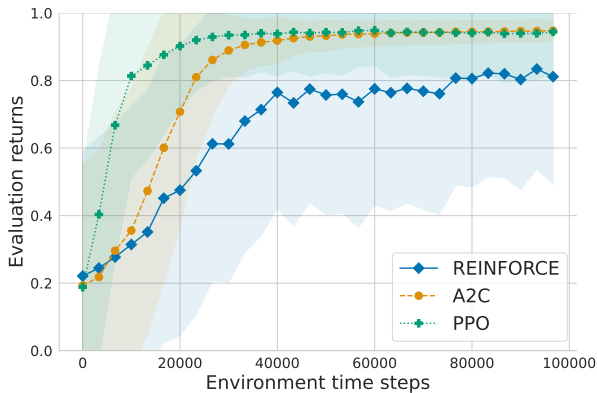
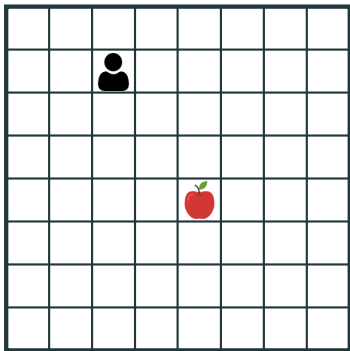
PPO Pseudocode

Algorithm Simplified proximal policy optimization (PPO)

```
1: Initialize actor network  $\pi$  with random parameters  $\phi$ 
2: Initialize critic network  $V$  with random parameters  $\theta$ 
3: for every episode do
4:   for time step  $t = 0, 1, 2, \dots$  do
5:     Observe current state  $s^t$ 
6:     Sample action  $a^t \sim \pi(\cdot \mid s^t; \phi)$ 
7:     Apply action  $a^t$ ; observe reward  $r^t$  and next state  $s^{t+1}$ 
8:      $\pi_\beta(a^t \mid s^t) \leftarrow \pi(a^t \mid s^t; \phi)$ 
9:     for epoch  $e = 1, \dots, N_e$  do
10:       $\rho(s^t, a^t) \leftarrow \pi(a^t \mid s^t; \phi) \div \pi_\beta(a^t \mid s^t)$ 
11:      if  $s^{t+1}$  is terminal then
12:        Advantage  $Adv(s^t, a^t) \leftarrow r^t - V(s^t; \theta)$ 
13:        Critic target  $y^t \leftarrow r^t$ 
14:      else
15:        Advantage  $Adv(s^t, a^t) \leftarrow r^t + \gamma V(s^{t+1}; \theta) - V(s^t; \theta)$ 
16:        Critic target  $y^t \leftarrow r^t + \gamma V(s^{t+1}; \theta)$ 
17:      Actor loss  $\mathcal{L}(\phi) \leftarrow -\min \left( \begin{array}{l} \rho(s^t, a^t) Adv(s^t, a^t), \\ \text{clip}(\rho(s^t, a^t), 1 - \epsilon, 1 + \epsilon) Adv(s^t, a^t) \end{array} \right)$ 
18:      Critic loss  $\mathcal{L}(\theta) \leftarrow (y^t - V(s^t; \theta))^2$ 
19:      Update parameters  $\phi$  by minimising the actor loss  $\mathcal{L}(\phi)$ 
20:      Update parameters  $\theta$  by minimising the critic loss  $\mathcal{L}(\theta)$ 
```

- PPO executes multiple epochs of updates!
- For first epoch, $\pi = \pi_\beta$
 $\Rightarrow \rho = 1$
- After first epoch, $\pi \neq \pi_\beta$
 $\Rightarrow$ needs ρ to correct for off-policy data

Policy Gradient Algorithms in Practice



Example: RL in LLMs

PPO is used to fine-tune LLMs. How do we map the RL problem onto the language space?

- **State:** The prompt, and all tokens of the completion which have been so far generated (partial completion).
- **Action:** Which token to generate next.
- **Transitions:** Appending the token generated to the current partial completion.
- **Terminal state:** When the completion ends in the End of Sequence (EoS) token.
- **Rewards:** No reward for any transition, except when the EoS token is produced.

Example: RL in LLMs

How do we obtain the reward?

1. Get Human Preferences

- Ask users to compare different responses generated by the LLM for the same input prompt ('Which answer is better?').

2. Train a Reward Model

- Use this human preference data to train a separate model called a reward model.
- The reward model learns to predict what kind of responses humans will prefer and it learns to score responses based on helpfulness, harmlessness, and alignment with human preferences.

3. Fine-tune the LLM with RL

- Use the reward model as the environment for our LLM agent. The LLM generates responses (actions), and the reward model scores these responses (provides rewards).
- In essence, we're training the LLM to produce text that our reward model (which learned from human preferences) thinks is good.

Example: RL in LLMs

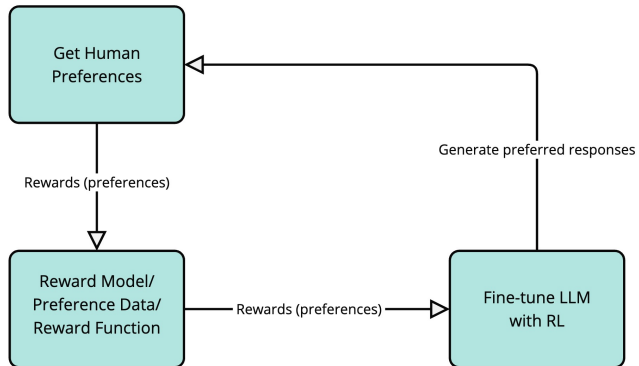


Figure: RL in LLMs.

Example: DeepSeek’s RL Algorithm

DeepSeek’s RL algorithm should now look at least a bit familiar!

2.2.1. Reinforcement Learning Algorithm

Group Relative Policy Optimization In order to save the training costs of RL, we adopt Group Relative Policy Optimization (GRPO) (Shao et al., 2024), which foregoes the critic model that is typically the same size as the policy model, and estimates the baseline from group scores instead. Specifically, for each question q , GRPO samples a group of outputs $\{o_1, o_2, \dots, o_G\}$ from the old policy $\pi_{\theta_{old}}$ and then optimizes the policy model π_{θ} by maximizing the following objective:

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(O|q)]$$
$$\frac{1}{G} \sum_{i=1}^G \left(\min \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)} A_i, \text{clip} \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1 - \epsilon, 1 + \epsilon \right) A_i \right) - \beta \mathbb{D}_{KL}(\pi_{\theta} || \pi_{ref}) \right), \quad (1)$$

$$\mathbb{D}_{KL}(\pi_{\theta} || \pi_{ref}) = \frac{\pi_{ref}(o_i|q)}{\pi_{\theta}(o_i|q)} - \log \frac{\pi_{ref}(o_i|q)}{\pi_{\theta}(o_i|q)} - 1, \quad (2)$$

where ϵ and β are hyper-parameters, and A_i is the advantage, computed using a group of rewards $\{r_1, r_2, \dots, r_G\}$ corresponding to the outputs within each group:

$$A_i = \frac{r_i - \text{mean}(\{r_1, r_2, \dots, r_G\})}{\text{std}(\{r_1, r_2, \dots, r_G\})}. \quad (3)$$

Figure: <https://arxiv.org/abs/2501.12948>

We covered:

- Deep Q-learning
- Moving-target problem and correlations of consecutive experiences
- Policy gradient algorithms
- Concurrent training of policies
- Observation, states, and histories under partial observability

Next we'll cover:

- Games as multi-agent interactions and solution concepts